Mälardalen University

PROJECT IN MATHEMATICS

On Permutation Groups

by

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MMA291 — Project in Mathematics

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MMA291 — Project in Mathematics

Date of presentation:
19 August 2026

Project name:
The Title of the Thesis

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Version: 1
12th August 2026

Supervisor:
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Examiner:
Peder Thompson

Comprising:
7.5 ECTS credits

Abstract

Under construction.

Acknowledgements

Thank you to Peder Thompson for guiding and encouraging me through the course of this project.

Contents

Chapter 1

Introduction

The motivation behind the research whose the results can be found largely within this work, was the desire to find an answer to a deceptively simple question: “What is the process by which methods for solving problems are found?”

The question arose within me in relation to a long-time fascination with the problem-solving process employed when finding methods for solving *twisty puzzles* – such as the *Rubik’s cube*. This focus led me to study *permutation groups*, and initially formulate my thoughts and findings in terms thereof.

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Chapter 2

Preliminaries

2.1 Groups

Definition 1 (Group)**.** In mathematics, a *group* G denotes a set of elements $\{a, b, c, \cdots\}$ and an accompanying binary operator $*$, together satisfying the following conditions:

1. *Closure*: For every ordered pair of elements in the group, the binary product thereof (as defined by the binary operator) is also an element of that group; $\forall a, b \in G \ \ (a * b \in G)$.

2. *Associativity*: The order wherein ordered pairs of elements are evaluated has no influence over the resulting product; $\forall a, b, c \in G \ \ ((a * b) * c = a * (b * c))$.

3. *Existence of Identity*: There exists in the group an *identity* element, the omission of which from any expression containing any other element or elements has no consequence on the evaluation of that expression; $\exists \varepsilon \in G \ \ (\forall a \in G \ (\varepsilon * a = a * \varepsilon = a))$.

4. *Existence of Inverse*: Each element in the group has its own unique *inverse*, who's product therewith evaluates to the identity; $\forall a \in G \ \ (\exists b \in G \ (a * b = \varepsilon))$, where ε is the identity.

Definition 2 (Commutativity)**.** Let G be a group. Then it is *commutative* (or *Abelian*) if $\forall a, b \in G \ (a * b = b * a)$.

Example 1. The integers $\mathbb{Z} = \{\cdots, -2, -1, 0, 1, 2, \cdots\}$ together with *addition* constitutes an Abelian group with 0 acting as its identity (since $\forall a \in \mathbb{Z} \ \ (0 + a = a + 0 = a)$). The negative numbers exist as the inverses of the positives, and vice versa. We know addition is associative, and since $\mathbb{Z}$ is infinite in each direction, we will never find a pair of elements therein whose sum is not also an element therein.

Definition 3 (Generation of a group)**.** A group G with the operator $*$ can be expressed by a set of *generators* $\mathscr{G} \subseteq G$ like so $G = (\langle \mathscr{G} \rangle, *)$ if and only if $\forall a \in G \ \left(\exists \mathbf{s} \in \bigcup_{n \in \mathbb{N}} \mathscr{G}^n \ \ (\mathbf{s}_1 * \cdots * \mathbf{s}_k = a) \right)$.

Definition 4 (Cyclicity)**.** A group which can be generated by a single element is called a *cyclic* group.

Example 2. The integers $\mathbb{Z}$ can be expressed as additively *generated* by the element 1, since 1 and its inverse (-1) together with addition $+$ can express the whole group: $(\mathbb{Z}, +) = (\langle\{1\}\rangle, +)$. Since the group $(\mathbb{Z}, +)$ can be generated by a single element, it is therefore a cyclic group. The group generated by 2 is the group of all *even* integers.

When the operator is given implicitly, a cyclic group G with the single generator $a \in G$ can be written as $G = \langle a \rangle$.

Definition 5 (Subgroup)**.** Let G be a group under a binary operator $*$. Then $H \subset G$ is a *subgroup* of G if H is also a group under $*$.

Example 3. The even integers $(\langle\{2\}\rangle, +)$ is a subgroup of the integers $(\mathbb{Z}, +)$ since it inherits associativity, existence of identity and existence of inverse from the integers, and is closed due to the fact that two even numbers can only produce another even number under addition.

Definition 6 (Coset)**.** Let G be a group under $*$ and $H \subset G$ be a subgroup thereof. Then a *left coset* aH of H in G is a set $\{a * h \mid h \in H\}$, where $a \in G$. Similarly, a *right coset* Ha is a set $\{h * a \mid h \in H\}$ where $a \in G$.

Lemma 1. *Let G be a finite group under $*$ and let $H \subset G$ be a subgroup thereof. Then $|aH| = |Ha| = |H|$.*

Proof. Let $H = \{h_1, h_2, \cdots, h_n\}$ and $a \in G$. Then $aH = \{a * h_1, a * h_2, \cdots, a * h_n\}$ with cardinality n, since – in a group – for any $h, h' \in H$, $a * h = a * h' \iff h = h'$, so each product is unique. Similarly, Ha also has cardinality $n = |H|$. □

Lemma 2. *Let G be a group under $*$ and let $H \subset G$ be a subgroup thereof. Then $h \in H \iff hH = Hh = H$.*

Proof. Let $h \in H$ and let $\mathbf{v}$ be any ordering of H – that is, any vector containing each element of H exactly once. Now, let $\mathbf{v}'$ also be such a vector defined by $\mathbf{v}'_i = h * \mathbf{v}_i$. Then for each index i, $h^{-1} * \mathbf{v}_i \in \mathbf{v}$ due to closure, whereby there exists an index j where $\mathbf{v}_j = h^{-1} * \mathbf{v}_i$, hence $\mathbf{v}_i = h * h^{-1} * \mathbf{v}_i = h * \mathbf{v}_j = \mathbf{v}'_j$. So $\mathbf{v}'$ is to $\mathbf{v}$ merely another ordering of H, wherefrom follows that $hH = H$.[1] A similar proof exists for the right-side case Hh. □

Lemma 3. *Let G be a finite group and let $H \subset G$ be a subgroup thereof. Then $|\{aH \mid a \in G\}| = |\{Ha \mid a \in G\}|$.*

Proof. For each pair $x, y \in G$ where $xH = yH$, there must be some $h \in H$ such that $xH = (yh)H = y(hH) = yH$ – that is, $h = y^{-1}x$. So $y^{-1}x \in H$, which means that $(y^{-1}x)^{-1} = yx^{-1} \in H$, since H is a group and thus has an inverse for each of its elements. Then $Hy^{-1} = H(yx^{-1})x^{-1} = Hx^{-1}$. So $xH = yH \iff Hx^{-1} = Hy^{-1}$ for any $x, y \in G$. Therefore, there are an equal number of unique cosets aH as there are unique cosets Ha for $a \in G$. □

Definition 7 (Index of a Subgroup)**.** Let H be a subgroup of a group G. Then the *index* $|G : H|$ of H in G is the number of unique left (or right) cosets of H in G – that is, $|\{aH \mid a \in G\}|$.

[1] There is, in fact, a more common and more straight-forward way of prooving this lemma. However, this method primes us for a way of thinking which will become relevant in the next section.

Theorem 1 (Lagrange's Theorem)**.** *If H is a subgroup of a finite group G under $*$, then* $|G : H| = \dfrac{|G|}{|H|}$.

Proof. Let $A = \{a_1, a_2, \cdots, a_n\} \subseteq G$ such that $G = a_1H \cup a_2H \cup \cdots \cup a_nH$ and $\forall a, b \in A \;\; (aH \cap bH = \emptyset)$. Then $|G| = |a_1H| + |a_2H| + \cdots + |a_nH|$ and thus $|G| = n \cdot |H|$, since $\forall a \in G \;\; (|aH| = |H|)$. Now, $n = |G : H|$, so by substitution we get $|G : H| = \dfrac{|G|}{|H|}$. □

Example 4. Let $G = (\langle\{1\}\rangle, +_8)$, where $+_8$ is addition modulo 8. Then $H = \{0, 4\}$ is a subgroup of G and the set of cosets of H in G is $\{aH \mid a \in \mathbb{Z}_8\} = \{aH \mid a \in \{0, 1, 2, 3\}\} = \{\{0, 4\}, \{1, 5\}, \{2, 6\}, \{3, 7\}\}$. Thus $|G : H| = 4 = \dfrac{8}{2} = \dfrac{|G|}{|H|}$.

2.2 Permutations

Definition 8 (Permutation)**.** A *permutation* π on a set A is a bijective map $\pi : A \to A$.

We write π^n for a permutation π to denote $\pi \circ \cdots \circ \pi$ where π is composed with itself $n \in \mathbb{N}$ times. π^{-n} denotes the composition $\pi^{-1} \circ \cdots \circ \pi^{-1}$ where the inverse π^{-1} is composed with itself $n \in \mathbb{N}$ times. The permutation π^0 always equals the identity.

Definition 9 (Permutation Group)**.** A *permutation group* over a set A is a group of permutations on A under composition.

Definition 10 (Support)**.** The *support* $\underline{\pi}$ of a permutation π on A is the set $\{a \in A \mid \pi(a) \neq a\}$.

A permutation whose support is the empty set equals the identity. We say that a permutation π on A *fixes* an element $a \in A$ if $a \notin \underline{\pi}$.

Definition 11 (Cycle)**.** A *cycle* is a permutation π such that $\forall a, b \in \underline{\pi} \;\; (\exists \phi \in \langle\pi\rangle \;\; (\phi(a) = b))$.

Since the support of the identity is the empty set, it follows that the identity also a cycle.

Permutations are commonly expressed using *cycle notation*, wherein each cycle is written as a collection of parenthesized series of elements. For example, a permutation π on $\{1, 2, 3, 4, 5\}$ defined by

$$\pi = \left\{ \begin{array}{c} 1 \mapsto 2 \\ 2 \mapsto 3 \\ 3 \mapsto 1 \\ 4 \mapsto 5 \\ 5 \mapsto 4 \end{array} \right\}$$

is expressed by its cycles like

$$\begin{array}{rcl} \pi & = & (1, 2, 3)(4, 5) \\ & = & (2, 3, 1)(5, 4) \\ & = & (4, 5)(3, 1, 2) \\ & \vdots & \end{array}$$

where each element is mapped to the succeeding element save for the last one, which is mapped to the first. When writing about permutations on elements 0 through 9, the commas in-between the numbers can be omitted like so: $(4, 6, 8, 2) = (4682)$.

Definition 12 (Set Form of a Permutation)**.** The *set form* $\widetilde{\pi}$ of a permutation π on A is the set of cycles on A with pairwise disjoint supports where $\forall a \in \underline{\pi}\ \ (\exists \phi \in \widetilde{\pi}\ \ (\phi(a) = \pi(a)))$ and $\bigcup\limits_{\phi \in \widetilde{\pi}} \underline{\phi} = \underline{\pi}$.

Example 5. Let $\pi = (12)(5349)(687)$. Then $\widetilde{\pi} = \{(12), (5349), (687)\}$.

Definition 13 (Permutation Form of a Set)**.** The *permutation form* $\mathrm{P}(S)$ of a set of cycles S on A (with pairwise disjoint supports) is the permutation on A whose set form is S.

When writing about the permutation form of a subset (or any other set relation) of the set form of a permutation, we use a shorthand: we write $\phi \subset \pi$ – for example – to mean $\phi = \mathrm{P}(S)$ where $S \subset \widetilde{\pi}$.

Example 6. Let $S = \{(12), (34), (56), (789)\}$. Then $\mathrm{P}(S) = (12)(34)(56)(789)$.

Definition 14 (Subpermutation)**.** The *subpermutation* $\overset{S}{\pi}$ of a permutation π for a set $S \subset \underline{\pi}$ is the permutation $\overset{S}{\pi} \subset \pi$ where $S \subseteq \underline{\overset{S}{\pi}}$.

A subpermutation which is a cycle is, conversely, called a *subcycle* – take for instance $\overset{\{a\}}{\pi} \in \pi$, where $a \in \underline{\pi}$.

Example 7. Let $\pi = (12)(34)(56)(789)$ and $\phi = (34)(789)$. Then $\phi \subset \pi$ and $\phi = \overset{\{3,7\}}{\pi} = \overset{\{2,3,7\}}{\pi} = \overset{\{2,3,7,9\}}{\pi} \cdots$.

Definition 15 (Orbit)**.** The *orbit* of an element $a \in A$ in a permutation π on A is the set $\underline{\overset{\{a\}}{\pi}}$.

Example 8. Let $\pi = (253)(798)$. Then the orbit of 8 in π is the set $\{7, 8, 9\}$.

Definition 16 (Transposition)**.** A *transposition* is a cycle π with $\left|\underline{\pi}\right| = 2$.

Example 9. Let $\pi = (19)(37)(58)$. Then each cycle in π is a transposition, since $\forall \phi \in \widetilde{\pi}\ \left(\left|\underline{\phi}\right| = 2\right)$.

Definition 17 (Order of a Permutation)**.** The *order* $|\pi|$ of a permutation π on A with $2 \leq \left|\underline{\pi}\right| < \infty$ is the smallest number $n \in \mathbb{N}$ (where $\mathbb{N} = \{1, 2, 3, \cdots\}$) such that $\pi^n = \varepsilon$, where $\varepsilon : A \longrightarrow A$ is the identity $\varepsilon(x) = x$.

Lemma 4. *Let ϕ be a cycle. Then the order of ϕ is $\left|\underline{\phi}\right|$.*

Proof. Let $a \in \underline{\phi}$. Then since $\left|\underline{\pi}\right| \geq 2$, $\phi(a) = b$ where $b \in \underline{\phi}$ and $a \neq b$. If $\phi(b) = a$, then $\underline{\phi} = \{a, b\}$ and the order is 2, since $\phi(\phi(x)) = \phi^2(x) = x$ for $x \in \{a, b\}$. Now, let $\alpha = (a_1, \cdots, a_k)$ have the order n and $\beta = (b_1, \cdots, b_k, b_{k+1})$ have the order m. Then $\alpha^{n-1}(a_1) = a_k$ and $\beta^{n-1}(b_1) = b_k$. So $\beta^n(b_1) = b_{k+1}$ and $\beta^{n+1}(b_1) = b_1$, wherefore $m = n + 1$. By induction, then, follows the lemma herefrom. □

Theorem 2. *Let π be a permutation where $\underline{\pi}$ is finite. Then the order of π is the least common multiple of the order of its cycles $\phi \in \widetilde{\pi}$.*

Proof. Let $\phi \in \widetilde{\pi}$ have the order n. Then $\phi^{kn} = \phi^n = \varepsilon$ for all $k \in \mathbb{N}$ (where ε is the identity), since $\phi^{kn} = \overbrace{\phi^n \circ \cdots \circ \phi^n}^{k} = \overbrace{\varepsilon \circ \cdots \circ \varepsilon}^{k} = \varepsilon$. Now, let $S = \{|\phi| \mid \phi \in \widetilde{\pi}\}$. Then $\forall \phi \in \widetilde{\pi} \left(\phi^{\mathrm{lcm}(S)} = \varepsilon\right)$, since $\exists k \in \mathbb{N}\ (k \cdot |\phi| = \mathrm{lcm}(S))$ for all $\phi \in \widetilde{\pi}$. Because every cycle in $\pi^{\mathrm{lcm}(S)}$ is the identity, it follows that $\pi^{\mathrm{lcm}(S)}$ is the identity also. The fact that $\mathrm{lcm}(S)$ is the *smallest* number with this property follows from the definition of the *least common multiple*. ☐

Example 10.

$$\begin{aligned} |(123)(45)| &= 6 \\ |(123)(456)(789)| &= 3 \\ |(012345)(6789)| &= 12 \end{aligned}$$

Definition 18 (Symmetric Group)**.** A *symmetric group G* over a set A is a permutation group over A where G is the set of all permutations on A.

Example 11. Let G be a symmetric group over $\{1, 2, 3\}$. Then

$$G = \{\varepsilon, (12), (13), (23), (123), (132)\}.$$

Symmetric groups over numbers 1 to n are often denoted S_n. (Thus the group described in Example 11 is the symmetric group S_3.)

Example 12. Let $G = S_4$ and $H = S_3$. Then H is a subgroup of G. The set of cosets $\{\pi H \mid \pi \in G\}$ evaluates to

$$\left\{ \begin{array}{l} \{(\varepsilon, (123), (23), (13), (132), (12)\} \\ \{(1324), (14)(23), (124), (1234), (134), (14)\} \\ \{(12)(34), (1243), (143), (1432), (34), (243)\} \\ \{(142), (1423), (13)(24), (24), (1342), (234)\} \end{array} \right\}$$

which has a cardinality of $\dfrac{|G|}{|H|} = \dfrac{4!}{3!} = \dfrac{24}{6} = 4$, as per Lagrange's theorem.

Theorem 3 (Cayley's Theorem)**.** *Let G be a group under $*$. Then G is isomorphic to a permutation group over G.*

Proof. Let $\lambda_g : G \longrightarrow G$ be defined by $\lambda_g(x) = g * x$ for any $g \in G$. Then λ_g is a permutation on G – namely

$$\lambda_g(x) = \left(\cdots, \lambda_{g^{-2}}(x), \lambda_{g^{-1}}(x), x, \lambda_g(x), \lambda_{g^2}(x), \cdots\right).$$

Now, let $G' = \left\{\lambda_g \,\middle|\, g \in G\right\}$. Then $(\lambda_a \circ \lambda_b)(x) = a * b * x = \lambda_{a*b}(x)$ for any $a, b, x \in G$. Since $a * b \in G$, the set G' is closed under composition. Since $\varepsilon \in G$, $\lambda_\varepsilon \in G'$ where

$\lambda_\varepsilon(x) = \varepsilon * x = x$ functions as the identity for G' under composition. G' is also associative under composition, since $(\lambda_a \circ \lambda_b) \circ \lambda_c = \lambda_{a*b} \circ \lambda_c = \lambda_{a*b*c} = \lambda_a \circ (\lambda_{b*c}) = \lambda_a \circ (\lambda_b \circ \lambda_c)$ for any $a, b \in G$. Also, if $a \in G$, then $a^{-1} \in G$. So if $\lambda_a \in G'$, then $\lambda_{a^{-1}} \in G'$ where $\lambda_{a^{-1}} \circ \lambda_a = \lambda_{a^{-1}*a} = \lambda_\varepsilon$ – or, in words, there is a unique inverse for every element in G' under composition. Since G' is a set of permutations and satisfies closure, associativity, existence-of-inverse and existence-of-inverses under composition, G' is is a permutation group under composition.

Lastly, let $\rho : G \to G'$ a bijective function defined by $\rho(g) = \lambda_g$. Then $\rho(a * b) = \lambda_{a*b} = \lambda_a \circ \lambda_b = \rho(a) \circ \rho(b)$. So any true expression $g_1 * \cdots * g_n = h$ where $g_1, \cdots, g_n, h \in G$ will hold for an equivalent expression $\rho(g_1) \circ \cdots \circ \rho(g_n) = \rho(h)$ and vice versa (using ρ^{-1}). Therefore, ρ is an isomorphism between G and G'. $\square$

Chapter 3

The Conjugates and Commutators of Permutations

Herein, all lowercase Greek letters are implied to be elements in a permutation group G on A. Elements of A are represented by lowercase Latin characters. Further, the operator $\circ$ will henceforth be implicit, and thus omitted from expressions. As an example, take $\pi \circ \phi = \pi\phi$.

Definition 19 (Conjugate)**.** The *conjugate* $\searrow^{\alpha}_{\beta}$ of β by α is the composition $\alpha^{-1}\beta\alpha$.

Definition 20 (Commutator)**.** The *commutator* $\nearrow\!\!\!\!\!\searrow^{\alpha}_{\beta}$ of α and β is the composition $\beta^{-1}\alpha^{-1}\beta\alpha$.

These two constructions – though they can seem arbitrary at first – have interesting and useful properties when working within groups – specifically when the supports of their two permutations share elements.

In the trivial case where they share no element, the permutations commute, so the conjugate of α and β is β, and the commutator thereof is the identity; $\alpha \cap \beta = \emptyset$ implies that $\alpha^{-1}\beta\alpha = \beta$ and $\beta^{-1}\alpha^{-1}\beta\alpha = e$.

Lemma 5. *Let* $\underline{\alpha} \cap \underline{\beta} = \{c\}$. *Then* $\overset{\{c\}}{\searrow^{\alpha}_{\beta}} = \left(\alpha^{-1}(c), \beta(c), \cdots, \beta^{-1}(c)\right)$.

Proof. For each $b \in \underline{\alpha} \setminus \left\{\alpha^{-1}(c)\right\}$, $\alpha(b) \notin \underline{\beta}$. Since $\beta\alpha(b) = \alpha(b)$, the conjugate can be rewritten for b: $\left(\alpha^{-1}\beta\alpha\right)(b) = \left(\alpha^{-1}\alpha\right)(b) = b$. In the case of $\alpha^{-1}(c)$, $\left(\alpha^{-1}\beta\alpha\alpha^{-1}\right)(c) = \left(\alpha^{-1}\beta\right)(c) = \beta(c)$, since $\beta(c) \notin \underline{\alpha} \implies \left(\alpha^{-1}\beta\right)(c) = \beta(c)$. For each $d \in \underline{\beta} \setminus \left\{\beta^{-1}(c), c\right\}$, $d, \beta(d) \notin \underline{\alpha}$, so $\left(\alpha^{-1}\beta\alpha\right)(d) = \beta(d)$. Lastly, $\beta^{-1}(c)$ maps to $\left(\alpha^{-1}\beta\alpha\beta^{-1}\right)(c) = \left(\alpha^{-1}\beta\beta^{-1}\right)(c) = \alpha^{-1}(c)$. This gives the function

$$\searrow^{\alpha}_{\beta}(x) = \begin{cases} x & : x \in \underline{\alpha} \setminus \left\{\alpha^{-1}(c)\right\} \\ \beta(c) & : x = \alpha^{-1}(c) \\ \beta(x) & : x \in \underline{\beta} \setminus \left\{\beta^{-1}(c), c\right\} \\ \alpha^{-1}(c) & : x = \beta^{-1}(c) \end{cases} \tag{3.1}$$

whose subcycle of c is written as $\left(\alpha^{-1}(c), \beta(c), \cdots, \beta^{-1}(c)\right)$. □

Corollary 1. *If β is a transposition, then so is $\searrow^{\alpha}_{\beta}$.*

Proof. If β is a transposition, then $\beta = (c, \beta(c))$ and $\beta(c) = \beta^{-1}(c)$. Using (3.1) we get

$$\searrow^{\alpha}_{\beta}\ (x) = \begin{cases} x & : x \in \underline{\alpha} \setminus \left\{\alpha^{-1}(c)\right\} \\ \beta(c) & : x = \alpha^{-1}(c) \\ \beta(x) & : x \in \emptyset \\ \alpha^{-1}(c) & : x = \beta^{-1}(c) \end{cases} \tag{3.2}$$

$$= \begin{cases} x & : x \in \underline{\alpha} \setminus \left\{\alpha^{-1}(c)\right\} \\ \beta(c) & : x = \alpha^{-1}(c) \\ \alpha^{-1}(c) & : x = \beta(c) \end{cases} \tag{3.3}$$

$$= \left(\alpha^{-1}(c), \beta(c)\right). \tag{3.4}$$

□

Lemma 6. *Let $\underline{\alpha} \cap \underline{\beta} = \{c\}$. Then $\nearrow\!\!\!\!\searrow^{\alpha}_{\beta} = \left(c, \beta^{-1}(c), \alpha^{-1}(c)\right)$.*

Proof. Using (3.1) from Lemma 5, we get

$$\nearrow\!\!\!\!\searrow^{\alpha}_{\beta}\ (x) = \left(\beta^{-1} \searrow^{\alpha}_{\beta}\right)(x) = \begin{cases} \left(\beta^{-1}\right)(x) & : x \in \underline{\alpha} \setminus \left\{\alpha^{-1}(c)\right\} \\ \left(\beta^{-1}\beta\right)(c) & : x = \alpha^{-1}(c) \\ \left(\beta^{-1}\beta\right)(x) & : x \in \underline{\beta} \setminus \left\{\beta^{-1}(c), c\right\} \\ \left(\beta^{-1}\alpha^{-1}\right)(c) & : x = \beta^{-1}(c) \end{cases} \tag{3.5}$$

$$= \begin{cases} x & : x \in \underline{\alpha} \setminus \left\{\alpha^{-1}(c), c\right\} \\ \beta^{-1}(x) & : x = c \\ c & : x = \alpha^{-1}(c) \\ x & : x \in \underline{\beta} \setminus \left\{\beta^{-1}(c), c\right\} \\ \alpha^{-1}(c) & : x = \beta^{-1}(c) \end{cases} \tag{3.6}$$

$$= \begin{cases} x & : x \in (\underline{\alpha} \cup \underline{\beta}) \setminus \left\{\alpha^{-1}(c), \beta^{-1}(c), c\right\} \\ c & : x = \alpha^{-1}(c) \\ \alpha^{-1}(c) & : x = \beta^{-1}(c) \\ \beta^{-1}(c) & : x = c \end{cases} \tag{3.7}$$

$$= \left(c, \beta^{-1}(c), \alpha^{-1}(c)\right). \tag{3.8}$$

□

Lemma 7. *Let $\underline{\alpha} \cap \underline{\beta} = \{c_1, \cdots, c_n\}$, $\alpha \neq \beta$ for $n \geq 2$ where $\alpha^{k-1}(c_1) = \beta^{k-1}(c_1) = c_k$ for $k \in \{1, \cdots, n\}$. Then ${\searrow}^{\overset{\{c\}}{\alpha}}_{\beta} = \left(\alpha^{-1}(c_1), c_1, \cdots, c_{n-1}, \beta(c_n), \cdots, \beta^{-1}(c_1)\right)$.*

Proof. Let $a \in \left\{c_n, \alpha(c_n), \cdots, \alpha^{-2}(c_1)\right\}$. Then $\alpha(a) \notin \underline{\beta}$, so $\left(\alpha^{-1}\beta\alpha\right)(a) = \left(\alpha^{-1}\alpha\right)(a) = a$. Conversely, each $b \in \left\{\beta(c_n), \cdots, \beta^{-2}(c_1)\right\}$ maps to $\left(\alpha^{-1}\beta\alpha\right)(b) = \beta(b)$, since $b, \beta(b) \notin \underline{\alpha}$. As for the elements $\{c_1, \cdots, c_{n-2}\}$, none thereof diverge from the common path $\alpha^{k-1}(c_1) =$

$\beta^{k-1}(c_1) = c_k$ for $k \in \{1, \cdots, n\}$, wherefore $c_j \overset{\alpha}{\mapsto} c_{j+1} \overset{\beta}{\mapsto} c_{j+2} \overset{\alpha^{-1}}{\mapsto} c_{j+1}$ in $\searrow^{\alpha}_{\beta}$ for each $j \in \{1, \cdots, n-2\}$. Lastly, $\alpha^{-1}(c_1)$ maps to $\left(\alpha^{-1}\beta\alpha\alpha^{-1}\right)(c_1) = \left(\alpha^{-1}\beta\right)(c_1) = \alpha^{-1}(c_2) = c_1$, $\beta^{-1}(c_1)$ maps to $\left(\alpha^{-1}\beta\alpha\beta^{-1}\right)(c_1) = \left(\alpha^{-1}\beta\beta^{-1}\right)(c_1) = \alpha^{-1}(c_1)$ and c_{n-1} maps to $\left(\alpha^{-1}\beta\alpha\right)(c_{n-1}) = \left(\alpha^{-1}\beta\right)(c_n) = \beta(c_n)$. This gives the function

$$\searrow^{\alpha}_{\beta}(x) = \begin{cases} x & : x \in \left\{c_n, \alpha(c_n), \cdots, \alpha^{-2}(c_1)\right\} \\ c_1 & : x = \alpha^{-1}(c_1) \\ \beta(x) & : x \in \left\{\beta(c_n), \cdots, \beta^{-2}(c_1)\right\} \cup \{c_1, \cdots, c_{n-2}\} \\ \beta(c_n) & : x = c_{n-1} \\ \alpha^{-1}(c_1) & : x = \beta^{-1}(c_1) \end{cases} \tag{3.9}$$

$$\implies \overset{\{c_1\}}{\searrow^{\alpha}_{\beta}} = \left(\alpha^{-1}(c_1), c_1, \cdots, c_{n-1}, \beta(c_n), \cdots, \beta^{-1}(c_1)\right). \tag{3.10}$$

□

Lemma 8. *Let α and β be permutations with the same properties as those in Lemma 7. Then* $\nearrow\!\!\!\!\!\searrow^{\alpha}_{\beta} = \left(\alpha^{-1}(c_1), \beta^{-1}(c_1)\right)(c_{n-1}, c_n)$.

Proof. From (3.9) from Lemma 7 we get

$$\nearrow\!\!\!\!\!\searrow^{\alpha}_{\beta}(x) = \begin{cases} \beta^{-1}(x) & : x \in \left\{c_n, \alpha(c_n), \cdots, \alpha^{-2}(c_1)\right\} \\ \beta^{-1}(c_1) & : x = \alpha^{-1}(c_1) \\ \left(\beta^{-1}\beta\right)(x) & : x \in \left\{\beta(c_n), \cdots, \beta^{-2}(c_1)\right\} \cup \{c_1, \cdots, c_{n-2}\} \\ \left(\beta^{-1}\beta\right)(c_n) & : x = c_{n-1} \\ \left(\beta^{-1}\alpha^{-1}\right)(c_1) & : x = \beta^{-1}(c_1) \end{cases} \tag{3.11}$$

$$= \begin{cases} x & : x \in \left\{\alpha(c_n), \cdots, \alpha^{-2}(c_1)\right\} \\ \beta^{-1}(c_n) & : x = c_n \\ \beta^{-1}(c_1) & : x = \alpha^{-1}(c_1) \\ x & : x \in \left\{\beta(c_n), \cdots, \beta^{-2}(c_1)\right\} \cup \{c_1, \cdots, c_{n-2}\} \\ c_n & : x = c_{n-1} \\ \alpha^{-1}(c_1) & : x = \beta^{-1}(c_1) \end{cases} \tag{3.12}$$

$$= \begin{cases} x & : x \in \left\{c_{n-1}, c_n, \alpha^{-1}(c_1), \beta^{-1}(c_1)\right\} \\ \beta^{-1}(c_1) & : x = \alpha^{-1}(c_1) \\ \alpha^{-1}(c_1) & : x = \beta^{-1}(c_1) \\ c_n & : x = c_{n-1} \\ c_{n-1} & : x = c_n \end{cases} \tag{3.13}$$

$$= \left(\alpha^{-1}(c_1), \beta^{-1}(c_1)\right)(c_{n-1}, c_n). \tag{3.14}$$

□

Lemma 9. *Let α and β be permutations such that* $\left\{\alpha(b) \,\middle|\, b \in \underline{\beta}\right\} \cap \underline{\beta} = \emptyset$ *and .*

DEFINE PERMUTATION *SHAPE*.

Lemma 10. *Let* $\emptyset \neq \underline{\alpha} \cap \underline{\beta} \neq \underline{\alpha} \cup \underline{\beta}$ *and* $\forall \beta' \in \beta \left(\underline{\beta'} \subseteq \underline{\alpha} \vee \underline{\beta'} \cap \underline{\alpha} = \emptyset\right)$. *Now, let* γ *be a permutation* $\gamma \subset \beta$ *such that* $\underline{\beta \setminus \gamma} \cap \underline{\alpha} = \emptyset$ *and* $\underline{\gamma} \subseteq \underline{\alpha}$. *Then* $\rtimes_{\beta}^{\alpha} = \rtimes_{\gamma}^{\alpha}$ *? Then the shape of* γ^{-1} *is applied on* $\underline{\gamma}$ *and* γ *is applied on* $\left\{\alpha(g) \,\middle|\, g \in \underline{\gamma}\right\}$ *by the commutator* $\rtimes_{\beta}^{\alpha}$ *.*

Proof. Let $\beta = \beta_1, \cdots, \beta_n \beta'_1, \cdots, \beta'_m$ and $\beta' = \beta'_1, \cdots, \beta'_m$. Then

$$\rtimes_{\beta}^{\alpha} = \begin{cases} x & : x \in \underline{\beta} \setminus \underline{\beta'} \\ \beta'(x) & : x \end{cases} .$$

$\square$

Chapter 4

The 2×3 Game

The "2×3 game" is a simple puzzle with six tiles in a grid of numbers and a set of allowed moves which permute them. The goal of the game is to sort the grid into a "solved" state from some scrambled state.

$$\begin{array}{|ccc|}\hline \mathbf{5} & \mathbf{4} & \mathbf{2} \\ \mathbf{3} & \mathbf{6} & \mathbf{1} \\ \hline \end{array} \xmapsto{\text{Move}_1} \cdots \xmapsto{\text{Move}_n} \begin{array}{|ccc|}\hline \mathbf{1} & \mathbf{2} & \mathbf{3} \\ \mathbf{4} & \mathbf{5} & \mathbf{6} \\ \hline \end{array}$$

Formally, the game can and will herein henceforth be modeled as follows. Let $G := (\langle \mathfrak{M} \rangle, \circ)$ be a group compositionally generated by the set of moves $\mathfrak{M} := \{(123), (456), (14)\}$, named $U = (123)$ for *upper row*, $L = (456)$ for *lower row* and $S = (14)$ for *side flip*. A *problem* is a permutation $p \in G$, and a *solution* thereto is a vector $\mathbf{s} \in \mathfrak{M}^*$ such that

$$\mathop{\bigcirc}\limits_{i=n}^{1} \mathbf{s}_i = \mathbf{s}_n \circ \mathbf{s}_{n-1} \circ \cdots \circ \mathbf{s}_1 = p$$

where n denotes the length of $\mathbf{s}$. Note that $\mathbf{s}$ may include the implicitly given inverses of the group generators. Hereafter, $\mathop{\bigcirc}\limits_{i=n}^{1} \mathbf{s}_i$ will be written as $\acute{\mathbf{s}}$.

This group equals the symmetric group S_6. MAKE THIS A THEOREM/LEMMA. AND MAKE THE FOLLOWING ITS PROOF: This fact can be proven in the following way. Each transposition between the upper and lower row can be found with the nested conjugate

$$\begin{aligned} T_1(x, y) &= \left(L^{-y+3+1}\right)^{-1} \left(\left(U^{-x+1}\right)^{-1} (S) \left(U^{-x+1}\right)\right) \left(L^{-y+3+1}\right) \\ &= L^{y-4} U^{x-1} S U^{-x+1} L^{-y+4} \\ &= L^{y-1} U^{x-1} S U^{-x+1} L^{-y+1}, \end{aligned}$$

where $x \in \{1, 2, 3\}$ and $y \in \{4, 5, 6\}$. As an example, let $p = (26)$. Then

$$\begin{aligned}
T_1(2,6) &= L^{6-1}U^{2-1}SU^{-2+1}L^{-6+1} \\
&= L^5U^1SU^{-1}L^{-5} \\
&= (456)^5(123)^1(14)(123)^{-1}(456)^{-5} \\
&= (465)(123)(14)(132)(456) \\
&= (465)(24)(456) \\
&= (2,6).
\end{aligned}$$

This gives – after some reductions – the solution $\mathbf{s} = \left(L, U^{-1}, S, U, L^{-1}\right)$ visualized below.

$$\boxed{\begin{matrix} \mathbf{1} & \mathbf{6} & \mathbf{3} \\ \mathbf{4} & \mathbf{5} & \mathbf{2} \end{matrix}} \xmapsto{L} \boxed{\begin{matrix} \mathbf{1} & \mathbf{6} & \mathbf{3} \\ \mathbf{2} & \mathbf{4} & \mathbf{5} \end{matrix}} \xmapsto{U^{-1}} \boxed{\begin{matrix} \mathbf{6} & \mathbf{3} & \mathbf{1} \\ \mathbf{2} & \mathbf{4} & \mathbf{5} \end{matrix}}$$

$$\xmapsto{S} \boxed{\begin{matrix} \mathbf{2} & \mathbf{3} & \mathbf{1} \\ \mathbf{6} & \mathbf{4} & \mathbf{5} \end{matrix}} \xmapsto{U} \boxed{\begin{matrix} \mathbf{1} & \mathbf{2} & \mathbf{3} \\ \mathbf{6} & \mathbf{4} & \mathbf{5} \end{matrix}} \xmapsto{L^{-1}} \boxed{\begin{matrix} \mathbf{1} & \mathbf{2} & \mathbf{3} \\ \mathbf{4} & \mathbf{5} & \mathbf{6} \end{matrix}}$$

Each transposition between elements on the same row can be found with the nested conjugate

$$\begin{aligned}
T_2(x,y,R) &= \Big((R^{-x+1})^{-1}(S)(R^{-x+1})\Big)^{-1}\Big((R^{-y+1})^{-1}(S)(R^{-y+1})\Big)\Big((R^{-x+1})^{-1}(S)(R^{-x+1})\Big) \\
&= (R^{x-1}SR^{-x+1})^{-1}(R^{y-1}SR^{-y+1})(R^{x-1}SR^{-x+1}) \\
&= (R^{-(-x+1)}S^{-1}R^{-(x-1)})(R^{y-1}SR^{-y+1})(R^{x-1}SR^{-x+1}) \\
&= (R^{x-1}SR^{-x+1})(R^{y-1}SR^{-y+1})(R^{x-1}SR^{-x+1}) \\
&= R^{x-1}SR^{-x+1}R^{y-1}SR^{-y+1}R^{x-1}SR^{-x+1} \\
&= R^{x-1}SR^{-x+y}SR^{x-y}SR^{-x+1}
\end{aligned}$$

where $x, y \in R$ and $R \in \mathfrak{M} \setminus \{S\}$. As an example, let $p = (23)$. Since 2 and 3 belong to the upper row, R is set to U:

$$\begin{aligned}
T_2(2,3,U) &= U^{2-1}SU^{-2+3}SU^{2-3}SU^{-2+1} \\
&= U^1SU^1SU^{-1}SU^{-1} \\
&= (123)(14)(123)(14)(123)^{-1}(14)(123)^{-1} \\
&= (123)(14)(123)(14)(132)(14)(132) \\
&= (123)(14)(24)(14)(132) \\
&= (123)(12)(132) \\
&= (23).
\end{aligned}$$

This gives the solution $\mathbf{s} = \left(U^{-1}, S, U^{-1}, S, U, S, U\right)$ visualized below.

$$\boxed{\begin{matrix}1&3&2\\4&5&6\end{matrix}} \xmapsto{U^{-1}} \boxed{\begin{matrix}3&2&1\\4&5&6\end{matrix}} \xmapsto{S} \boxed{\begin{matrix}4&2&1\\3&5&6\end{matrix}} \xmapsto{U^{-1}} \boxed{\begin{matrix}2&1&4\\3&5&6\end{matrix}}$$

$$\xmapsto{S} \boxed{\begin{matrix}3&1&4\\2&5&6\end{matrix}} \xmapsto{U} \boxed{\begin{matrix}4&3&1\\2&5&6\end{matrix}} \xmapsto{S} \boxed{\begin{matrix}2&3&1\\4&5&6\end{matrix}} \xmapsto{U} \boxed{\begin{matrix}1&2&3\\4&5&6\end{matrix}}$$

Since a transposition of any two elements herein can be composed using the generators in $\mathfrak{M}$ – in other words, since the group G includes every transposition of the elements 1 through 6 – it follows that the group must include every permutation on the six elements, and thereby equal S_6.

Since G is the symmetric group S_6, there must exist a map $T_1(x, y) = (x, y)$, where $x \in \{1, 2, 3\}$ and $y \in \{4, 5, 6\}$.

From Corrolary 1, we know that since $\zeta_1(x, y)$ is a transposition, then $\zeta_1 = {\searrow}^{\alpha_1}_{\beta_1}$ where $\beta_1 = (x, c_1)$ and $\alpha_1(y) = c_1$. Similarly, from $\left|\underline{\beta_1}\right| = 2$ follows that $\beta_1 = {\searrow}^{\alpha_2}_{\beta_2}$ where $\beta_2 = (c_1, c_2)$ and $\alpha_2(x) = c_2$. Now, from $\mathfrak{M}$ we can find permutations matching these properties

$$T_1(x, y) = {\searrow}^{L^{-y+3+1}}_{{\searrow}^{U^{-x+1}}_{S}}$$

$$\begin{aligned} T_2(x, y, R) &= \left((R^{-x+1})^{-1}(S)(R^{-x+1})\right)^{-1}\left((R^{-y+1})^{-1}(S)(R^{-y+1})\right)\left((R^{-x+1})^{-1}(S)(R^{-x+1})\right) \\ &= R^{x-1}SR^{-x+y}SR^{x-y}SR^{-x+1} \end{aligned}$$

Since G is the symmetric group S_6, there must be a map $\zeta : \{1, \cdots, 6\}^2 \longrightarrow \mathcal{P}(\mathfrak{M}^{<\omega})$, where $\mathfrak{M}^{<\omega} = \bigcup_{n\in\mathbb{N}} \mathfrak{M}^n$ and $\mathbf{s} \in \zeta(x,y) \implies \acute{\mathbf{s}} = (x,y)$. In other words: there must exist a set of solutions to each transposition in S_6.

Since G is the symmetric group S_6, there must be an infinite[1] set of maps $\{1, \cdots, 6\}^2 \rightarrow \mathfrak{M}^{<\omega}$, where $\mathfrak{M}^{<\omega} = \bigcup_{n\in\mathbb{N}} \mathfrak{M}^n$ and $\mathbf{s} \in \zeta(x,y) \implies \acute{\mathbf{s}} = (x,y)$ where ζ is such a map.

There is a set of solutions to T_1, and α_1 and β_1 and α_2 and β_2. the solution is the intersection thereof. At each level there is a unique map (for that level) to a set of solutions[2]. It's recursive!

—

The lemmas and corollaries in chapter 3 – and indeed, any other observation, general or particular (about permutations or a specific game) – together construct a formal language.

Example: π where $|\underline{\pi}| = 2$ is a *pattern*, which can be substituted by another pattern: a conjugation $\searrow_{\beta_1}^{\alpha_1}$ where $\beta_1 = (x, c_1)$ and $\alpha_1(y) = c_1$ (as in the example above). The permutation $\beta_1 = (x, c_1)$ is, then, also a pattern to be substituted and so on until you have a permutation constructed from available moves ($\mathfrak{M}$?). You don't even need to evaluate them all if you know, for example, that ζ_1/T_1 exists.

All the lemmas and corollaries are in fact rewritings (with rewriting rules – in this case, formal math) ...

A *pattern* is a set p. If a set, vector, element (or other data) Ω matches the pattern p, then $\Omega \in p$. The set $\{p \mid \Omega \in p\}$ is the set of patterns to which some data Ω belongs.

Example

Let $a \gg b = [a_1, \cdots, a_n, b_1, \cdots, b_m]$, where $a = [a_1, \cdots, a_n]$ and $b = [b_1, \cdots, b_m]$.

$$\begin{array}{lcl}
\mathfrak{E} & := & \mathfrak{T} \cup \{\mathfrak{e}_1 \gg \mathfrak{o} \gg \mathfrak{e}_2 \mid \mathfrak{o} \in \{[\text{"+"}], [\text{"−"}]\} \wedge \mathfrak{e}_1, \mathfrak{e}_2 \in \mathfrak{E}\} \\
\mathfrak{T} & := & \mathfrak{F} \cup \{\mathfrak{t}_1 \gg \mathfrak{o} \gg \mathfrak{t}_2 \mid \mathfrak{o} \in \{[\text{"×"}], [\text{"÷"}]\} \wedge \mathfrak{t}_1, \mathfrak{t}_2 \in \mathfrak{T}\} \\
\mathfrak{F} & := & \mathfrak{N} \cup \{[\text{"("}] \gg \mathfrak{e} \gg [\text{")"}] \mid \mathfrak{e} \in \mathfrak{E}\} \\
\mathfrak{N} & := & \mathcal{N} \cup \mathcal{N}^* \\
\mathcal{N} & := & \{\text{"0"}, \cdots, \text{"9"}\}
\end{array}$$

[1]It is easy to see that this set is infinite, since one can append any solution with a permutation and its inverse to get a new solution. Take the solution $\mathbf{s}_2 = (\alpha, \beta, \gamma)$ as an example, then $\acute{\mathbf{s}}_1 = \acute{\mathbf{s}}_2$ where $\mathbf{s}_2 = (\alpha, \beta, \gamma, \alpha, \alpha^{-1})$.

[2]These are *patterns* with rewriting rules!

Chapter 5

Finding Methods

Definition 21 (Grimoire)**.** A *grimoire* Φ of a semigroup G, a set of *spells* $\mathfrak{S}_\Phi \subseteq G$ and a set of *movements* $\mathfrak{M} \subseteq \mathfrak{S}_\Phi$ is a map in $\mathfrak{S}_\Phi \to \mathcal{P}\left(\mathfrak{S}_\Phi^{<\omega}\right)$ such that $\mathfrak{s}_1 * \cdots * \mathfrak{s}_n = \mathfrak{s}$ – where $*$ is the binary operator belonging to G – for each $[\mathfrak{s}_1, \cdots, \mathfrak{s}_n] \in \Phi(\mathfrak{s})$ and $\mathfrak{s} \in \mathfrak{S}_\Phi \setminus \mathfrak{M}$.

Definition 22 (Exploration Function)**.** Let a map $\triangle : \boxed{\Phi} \times G^{<\omega} \to \boxed{\Phi}$ – where $\boxed{\Phi}$ is the set of all grimoires of a subgroup G – be defined for some grimoire Φ by $\forall \mathfrak{s} \in \mathfrak{S}_\Phi \setminus \{\mathfrak{s}_1 * \cdots * \mathfrak{s}_n\}$ $(\Phi'(\mathfrak{s}) = \Phi(\mathfrak{s}))$ and $\Phi'(\mathfrak{s}_1 * \cdots * \mathfrak{s}_n) = \Phi(\mathfrak{s}_1 * \cdots * \mathfrak{s}_n) \cup \{[\mathfrak{s}_1, \cdots, \mathfrak{s}_n]\}$!!!MAKE MORE CLEAR WHAT IS INCLUDED IN THE "IF" STATEMENT ("AND"???)!!!if $\mathfrak{s}_1 * \cdots * \mathfrak{s}_n \in \mathfrak{S}_\Phi$ and $\Phi'(\mathfrak{s}_1 * \cdots * \mathfrak{s}_n) = \{[\mathfrak{s}_1, \cdots, \mathfrak{s}_n]\}$ if not – where $\Phi' = \triangle(\Phi, [\mathfrak{s}_1, \cdots, \mathfrak{s}_n])$ and $\mathfrak{s}_1, \cdots, \mathfrak{s}_n \in \mathfrak{S}_\Phi$. (Note that since $\mathfrak{S}_\Phi = \mathrm{dom}(\Phi)$, then $\mathfrak{S}_{\Phi'} = \mathfrak{S}_\Phi \cup \{\mathfrak{s}_1 * \cdots * \mathfrak{s}_n\}$ if $\Phi' = \triangle(\Phi, [\mathfrak{s}_1, \cdots, \mathfrak{s}_n])$.) An *exploration function* $\mathcal{E}$ in G is a map in $\boxed{\Phi} \times \boxed{C}^{<\omega} \to \boxed{\Phi}$ defined by

$$\mathcal{E}(\Phi, [C_1, \cdots, C_n]) := \bigodot_{i=1}^{n} \triangle\left(\Phi, C_i\left(\mathfrak{S}_\Phi\right)\right),$$

!!!THE EXPLORATION IS NON-RECURSIVE!!!where $\boxed{C} \subset \mathcal{P}(G) \to G^{<\omega}$ and !!!THE DEFINITION OF C IS SLOPPY!!!$C(S) \in S^{<\omega}$ for each $C \in \boxed{C}$ and $S \in \mathcal{P}(G)$.

Definition 23 (Expansion Function)**.** An *expansion function* ʑ for a grimoire Φ of a semigroup G is a map in $\boxed{\Phi} \times \mathfrak{S}_\Phi \times \boxed{\Pi} \to \mathfrak{M}^{<\omega}$ – where $\boxed{\Phi}$ is the set of all grimoires of G – defined by

$$\text{ʑ}(\Phi, \mathfrak{s}, \Pi) := \begin{cases} \mathfrak{s} & : \text{ʑ}_{\text{step}}(\Phi, \mathfrak{s}, \Pi) = \mathfrak{s} \\ \text{ʑ}\left(\Phi, \text{ʑ}_{\text{step}}(\Phi, \mathfrak{s}, \Pi), \Pi\right) & : \text{otherwise} \end{cases}$$

where $\text{ʑ}_{\text{step}} : \boxed{\Phi} \times \mathfrak{S}_\Phi \times \boxed{\Pi} \to \mathfrak{S}_\Phi^{<\omega}$ is defined by

$$\text{ʑ}_{\text{step}}(\Phi, \mathfrak{s}, \Pi) := \sum_{i=1}^{|\mathfrak{s}|}\!\!\!\!\!\!\gg \left(\begin{cases} \mathfrak{s}_i & : \mathfrak{s}_i \in \mathfrak{M} \\ \Pi\left(\Phi\left(\mathfrak{s}_i\right)\right) & : \text{otherwise} \end{cases}\right),$$

where $\boxed{\Pi} \subset \mathcal{P}\left(G^{<\omega}\right) \to G^{<\omega}$ is a set of maps wherein $\Pi(S) \in S$ holds for each $\Pi \in \boxed{\Pi}$ and $S \in \mathcal{P}\left(G^{<\omega}\right)$.

Prove that grimoire-exploration with $\mathscr{E}$ or $\triangle$ is a semigroup.

Är gruppen definierad av *likheten* mellan en mängd utforskningar/utökningar av trollformelboken?